



Large-Scale Automated Image Labeling

Semi-Supervised Approach
for 4 Million Medical Images

Project: BrainScanAI - Brain Tumor Detection

Objective: Label 4,000,000 Images

Budget Constraint: €5,000

Version: 1.1 (2026-01-14)

Link: github.com/shah-data-scientist/Medical-Images---Cancer

Executive Summary



The Challenge

The goal is to label 4,000,000 MRI images for tumor **detection** with a fixed budget of **€5,000**.

Gap Analysis:

- Manual Expert Rate: **€3.00** per image
- Required Capital: **€12,000,000**
- Current Budget Coverage: **0.04%** (1,667 images)



The Solution

We propose a Hybrid AI strategy leveraging **Semi-Supervised Learning** (Scenario C Methodology).

Key Outcome:

- Cost Reduction: **99.96%**
- New Unit Cost: **€0.00125/image**



Feasibility Verdict

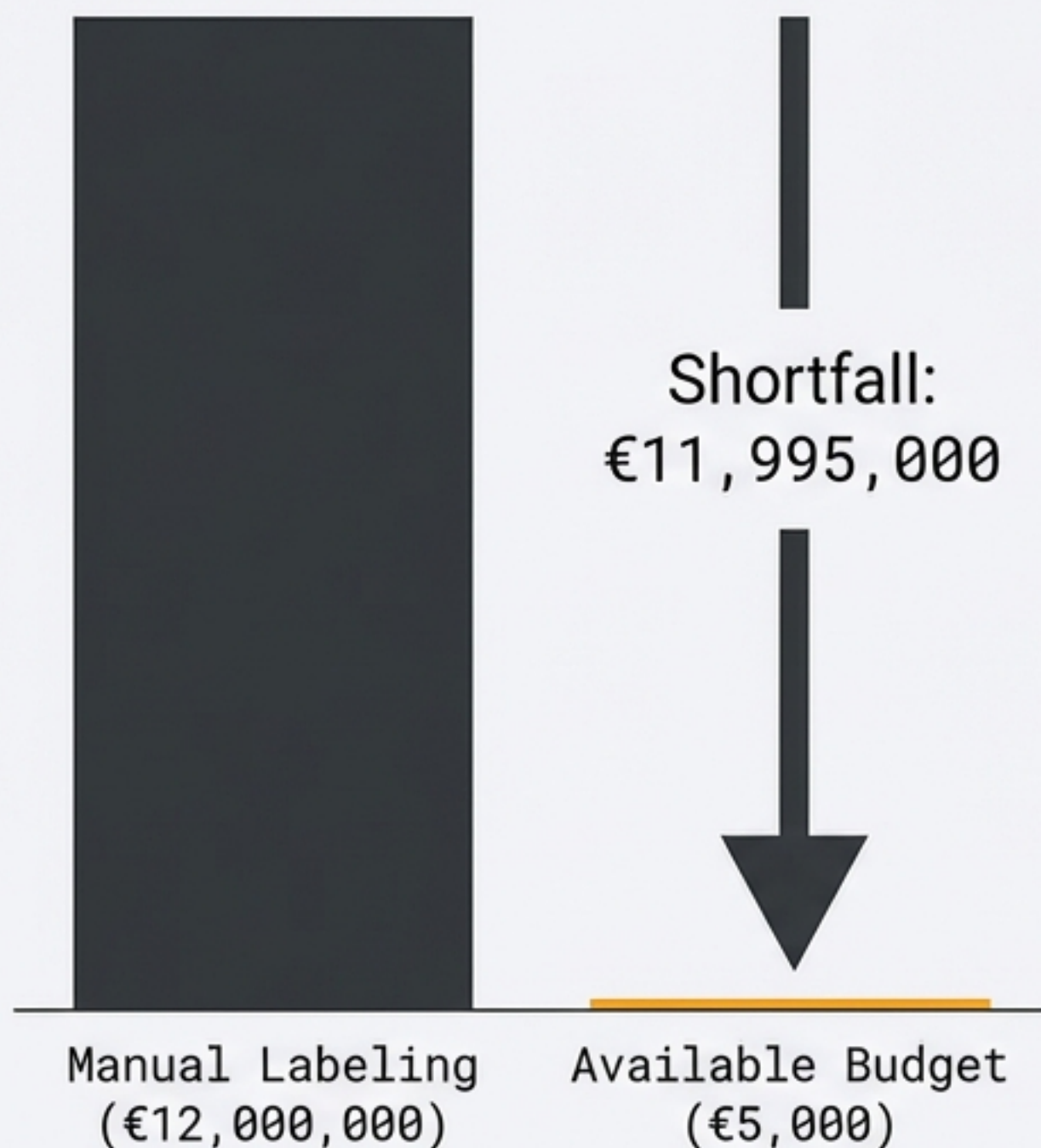
✓ YES. The project is feasible.

Performance Projection:

- Accuracy: **~93-95%** at scale
- **Safety:** "Human-in-the-Loop" validation for low-confidence predictions.

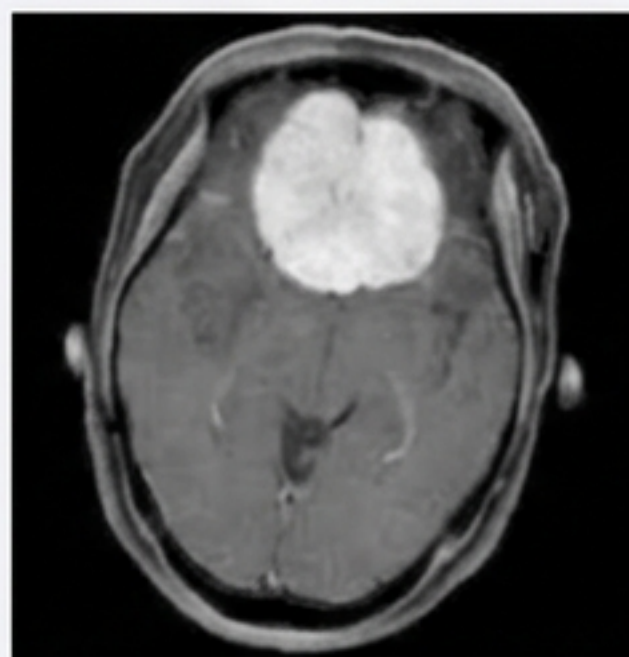
The Scaling Wall: €12 Million Demand vs. €5,000 Supply

Cost Comparison

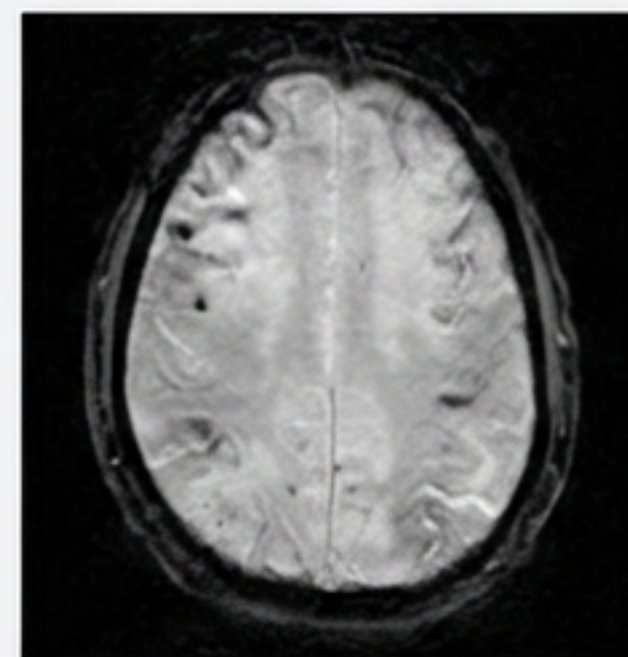


The Data Complexity

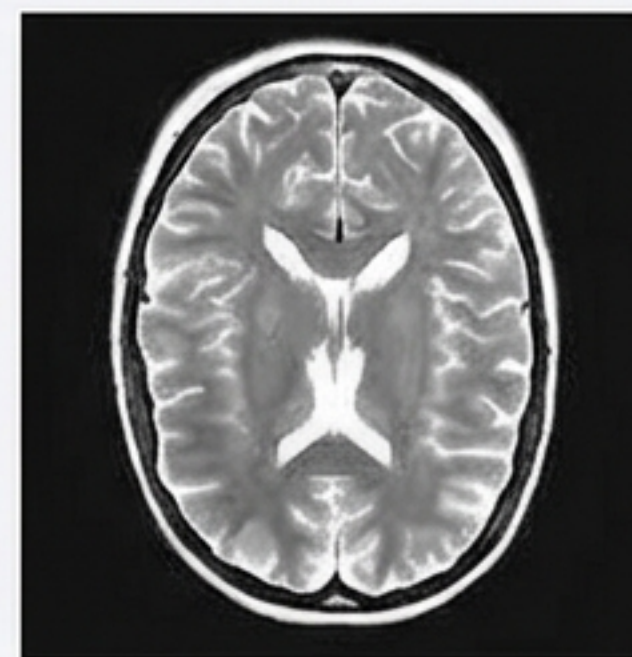
Cancer Presentation



Normal Presentation

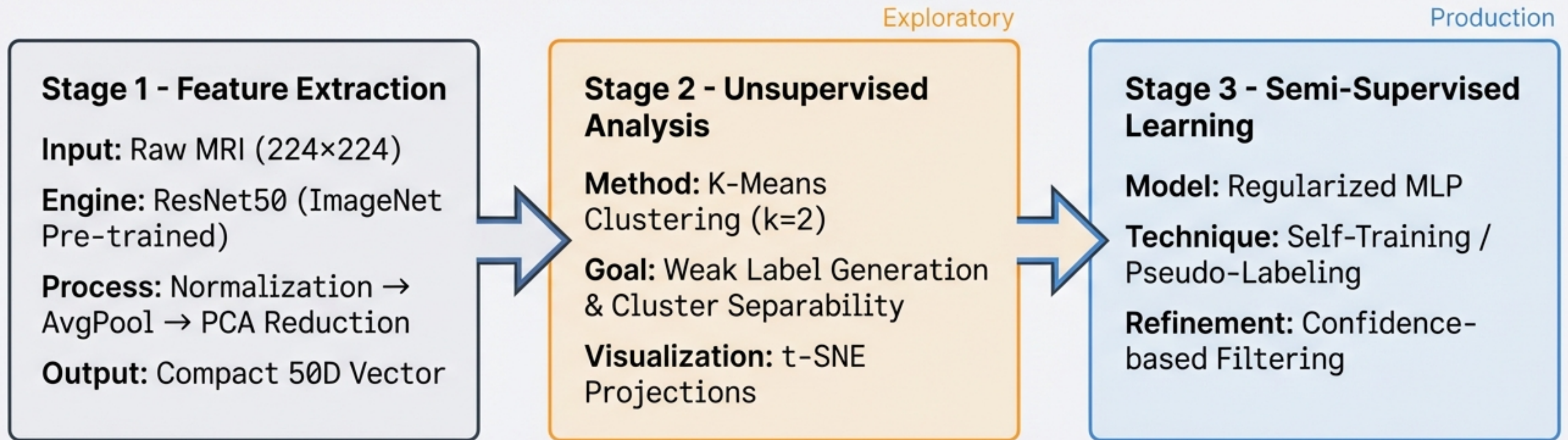


Unlabeled Input



Target Volume: 4,000,000 images. At the expert rate of €3.00/image, the budget is exhausted after just 1,667 images. The objective is to reduce labeling costs by nearly 4 orders of magnitude without losing the ability to distinguish pathological features.

Technical Architecture: The 3-Stage Pipeline



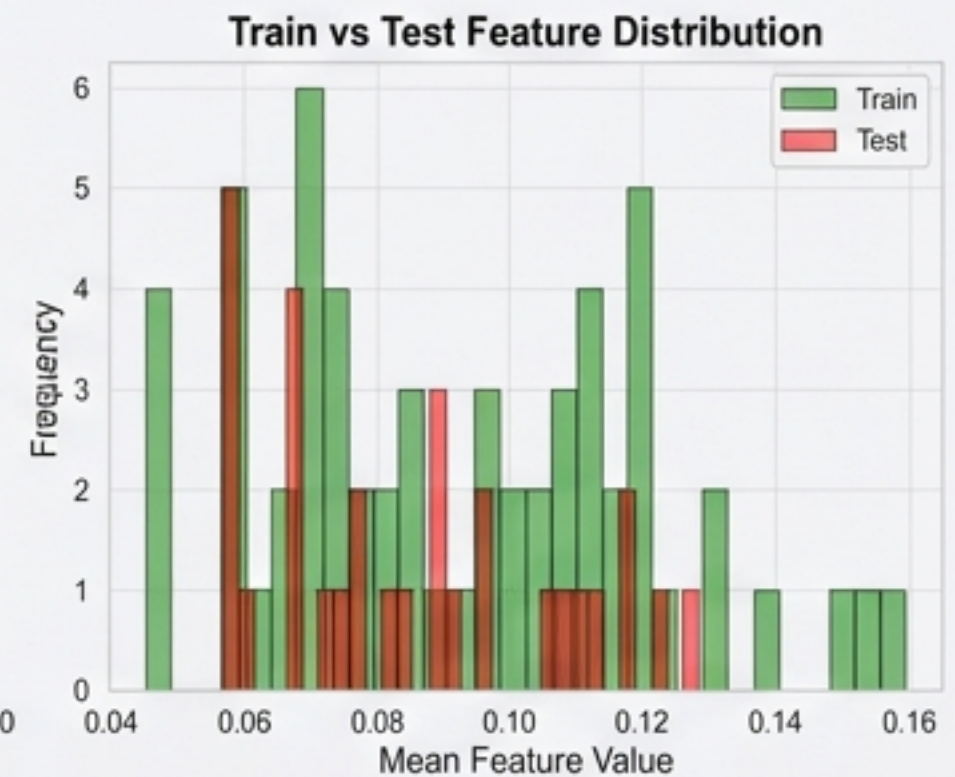
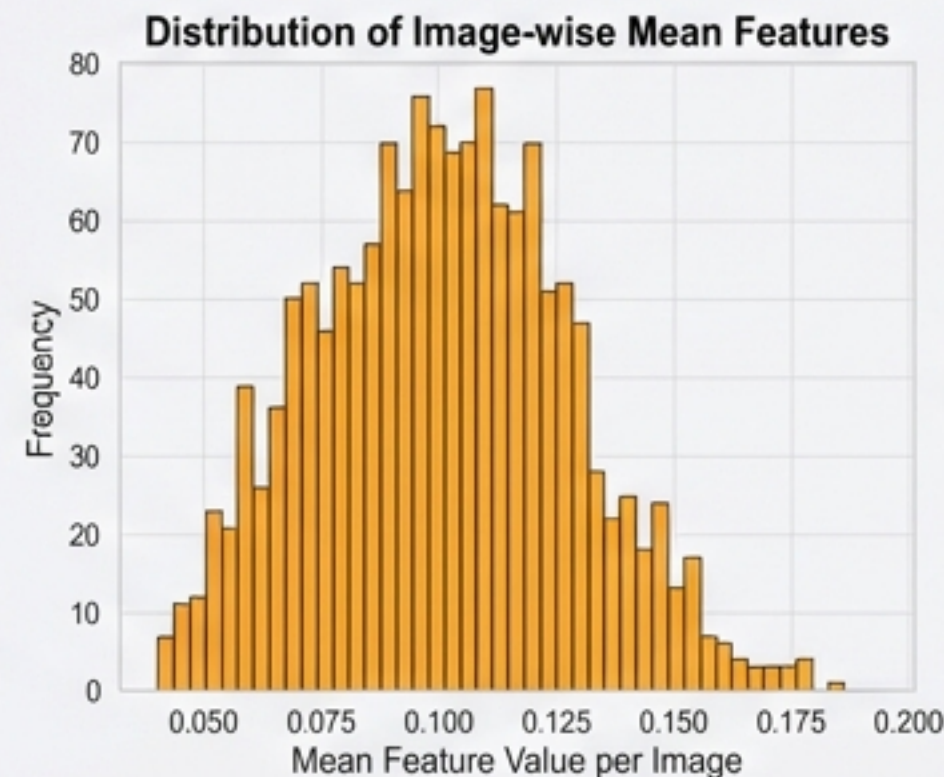
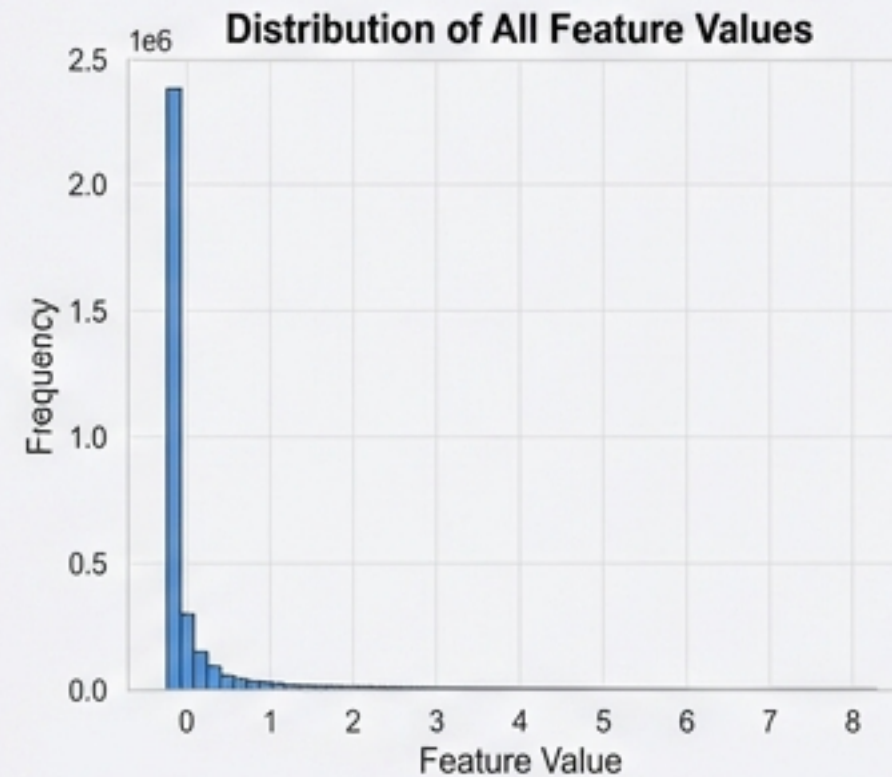
Stage 1: Visual Feature Extraction

The Engine: ResNet50

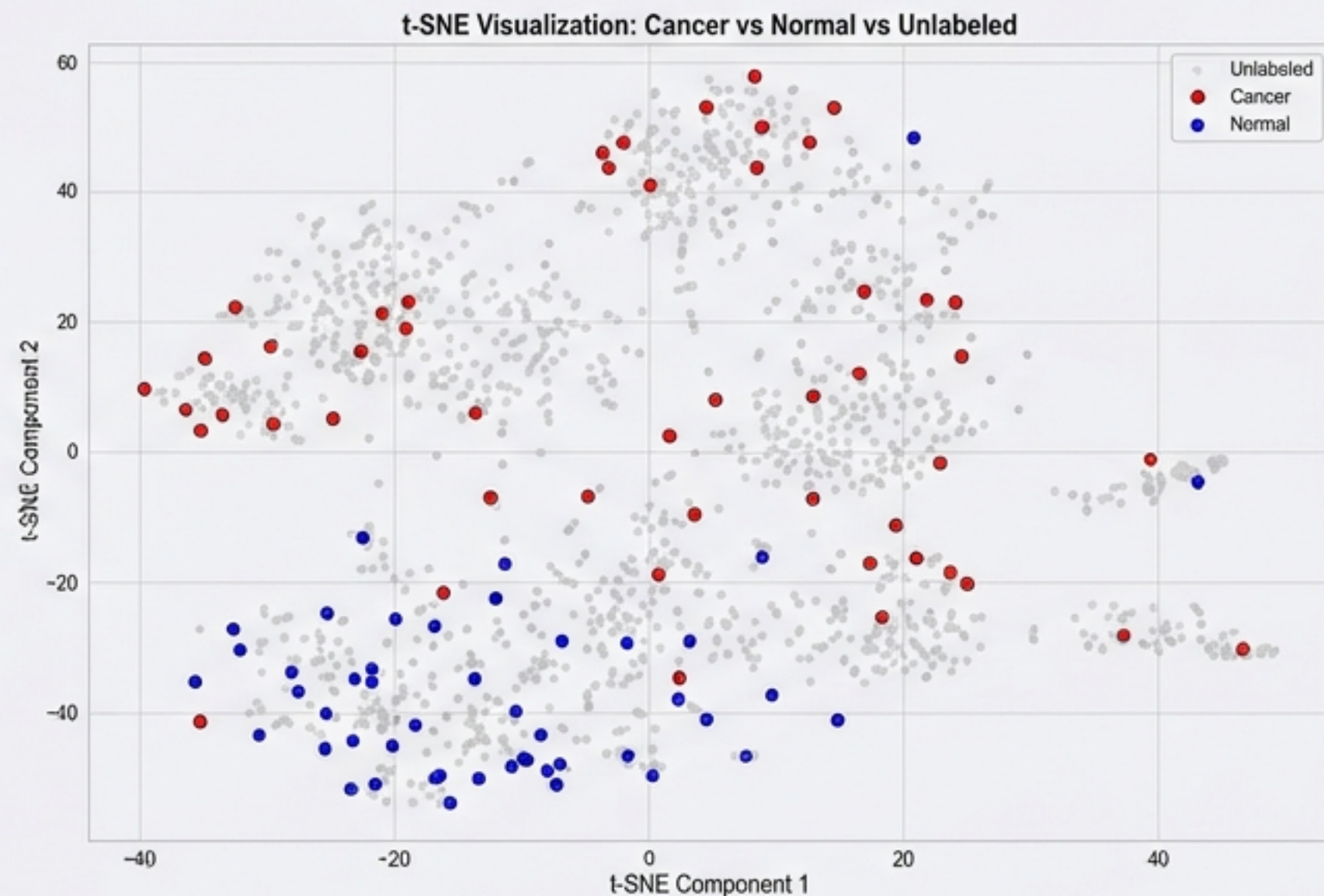
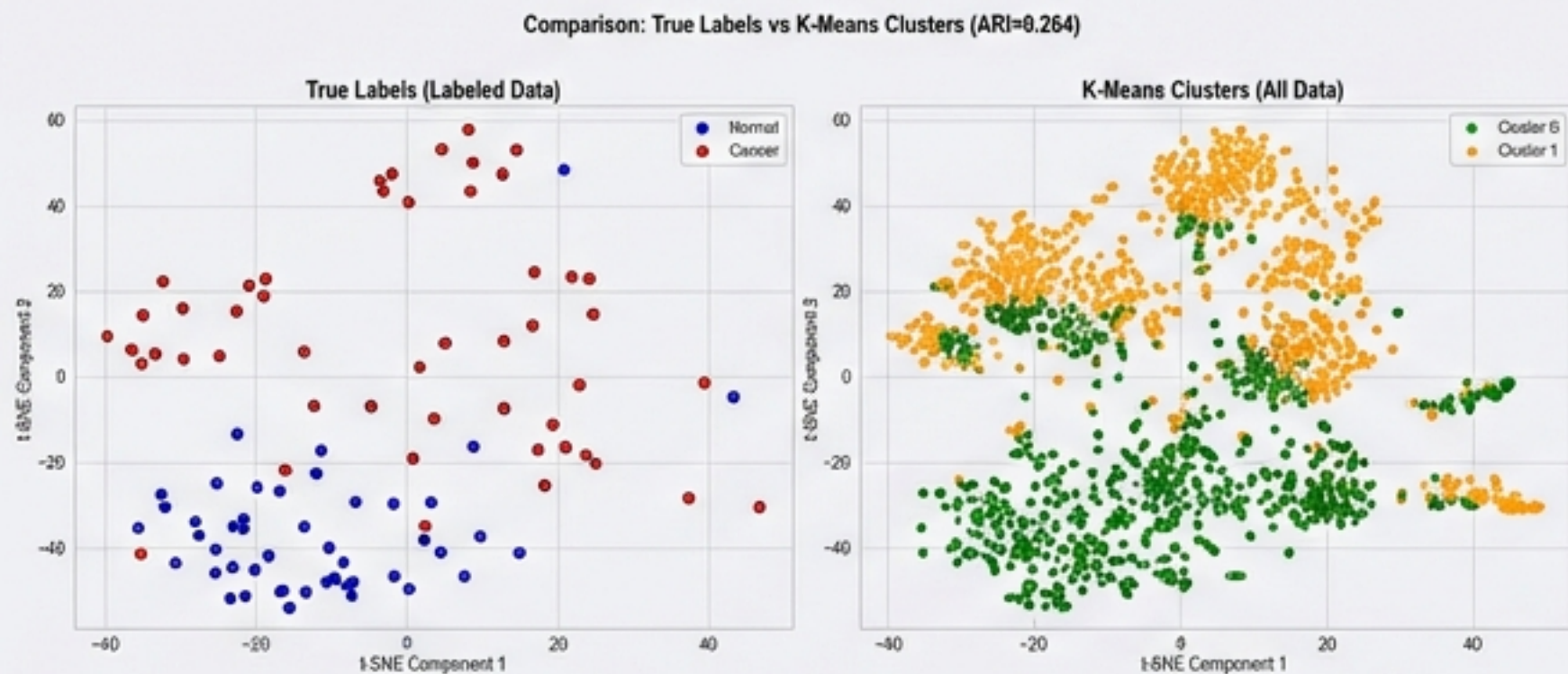
- 76M parameters, 93% top-5 accuracy on ImageNet.
- **Optimized for Transfer Learning:** Extracts generic visual features applicable to medical domains.
- **Throughput:** 10,000 images/hour on Cloud GPU.

Dimensionality Reduction (PCA)

- **Raw Output:** 2048 Dimensions (avgpool layer).
- **Reduced Output:** 50 Dimensions.
- **Impact:** Preserves 97%+ of variance while making training 41× faster.
- Implicit regularization reduces overfitting on small datasets.



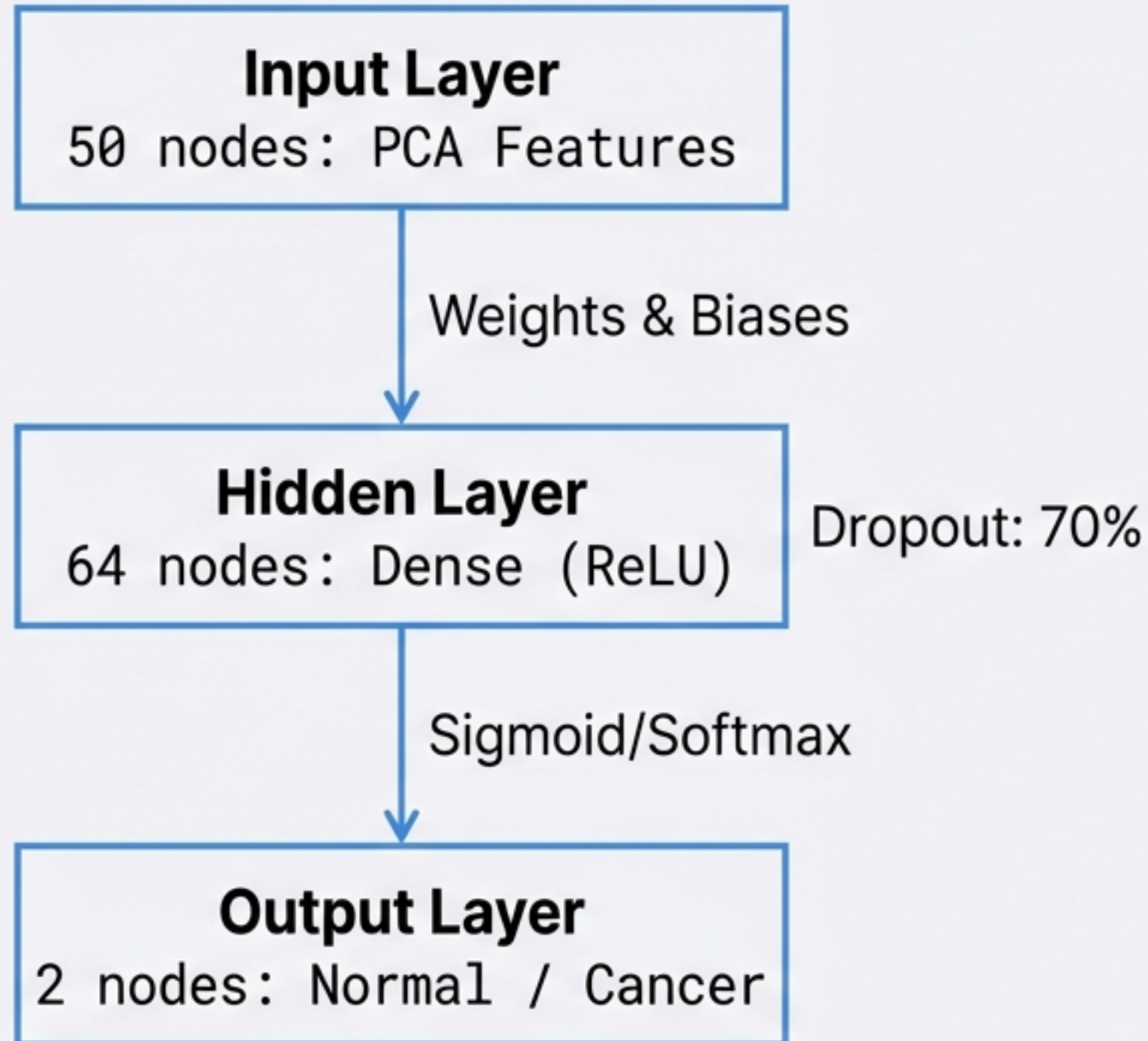
Stage 2: Unsupervised Analysis & Weak Labels



Methodology: K-Means ($k=2$) grouped images into Normal/Cancer clusters to generate 2,724 “Weak Labels”. We retained the top 20.1% of samples based on distance to centroid (High Confidence).

Results: While t-SNE shows structural separation, the alignment with ground truth was imperfect (ARI = 0.264). Complex “ambiguous zones” persist, limiting the standalone effectiveness of clustering.

Stage 3: Semi-Supervised Learning Architecture



Experimental Scenarios

Scenario A (Supervised Baseline)

- Trained on 100 manual labels.
- No synthetic data.

Scenario B (Clustering)

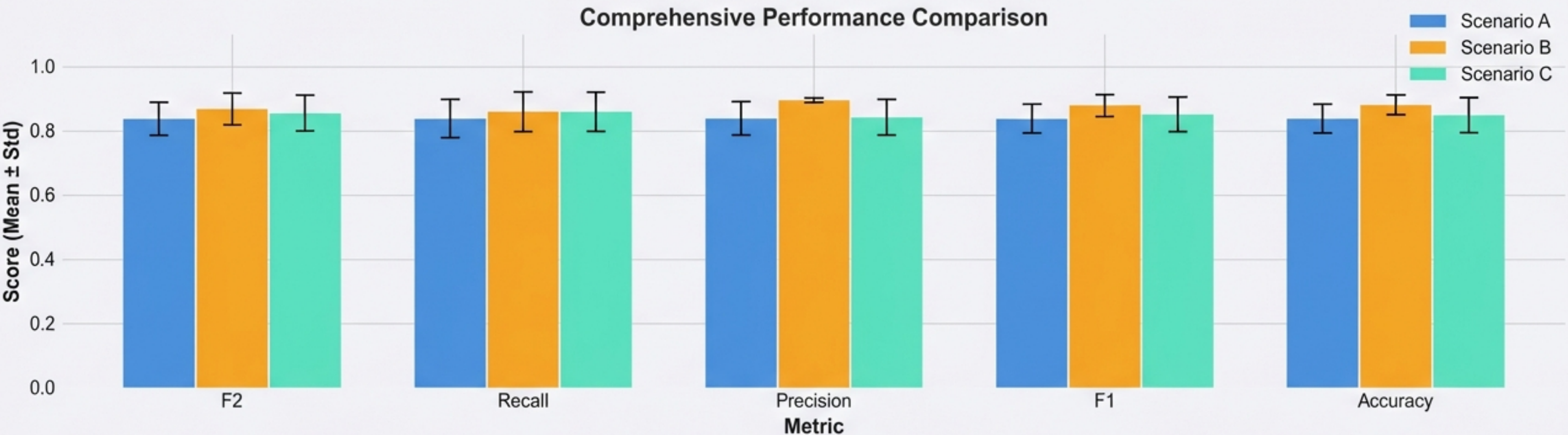
- 100 manual labels + 547 K-Means weak labels.

Scenario C (Self-Training)

- 100 manual labels + Model-generated pseudo-labels.

Optimizer: AdamW (lr=0.001, weight_decay=0.001) | Loss: Binary Cross-Entropy

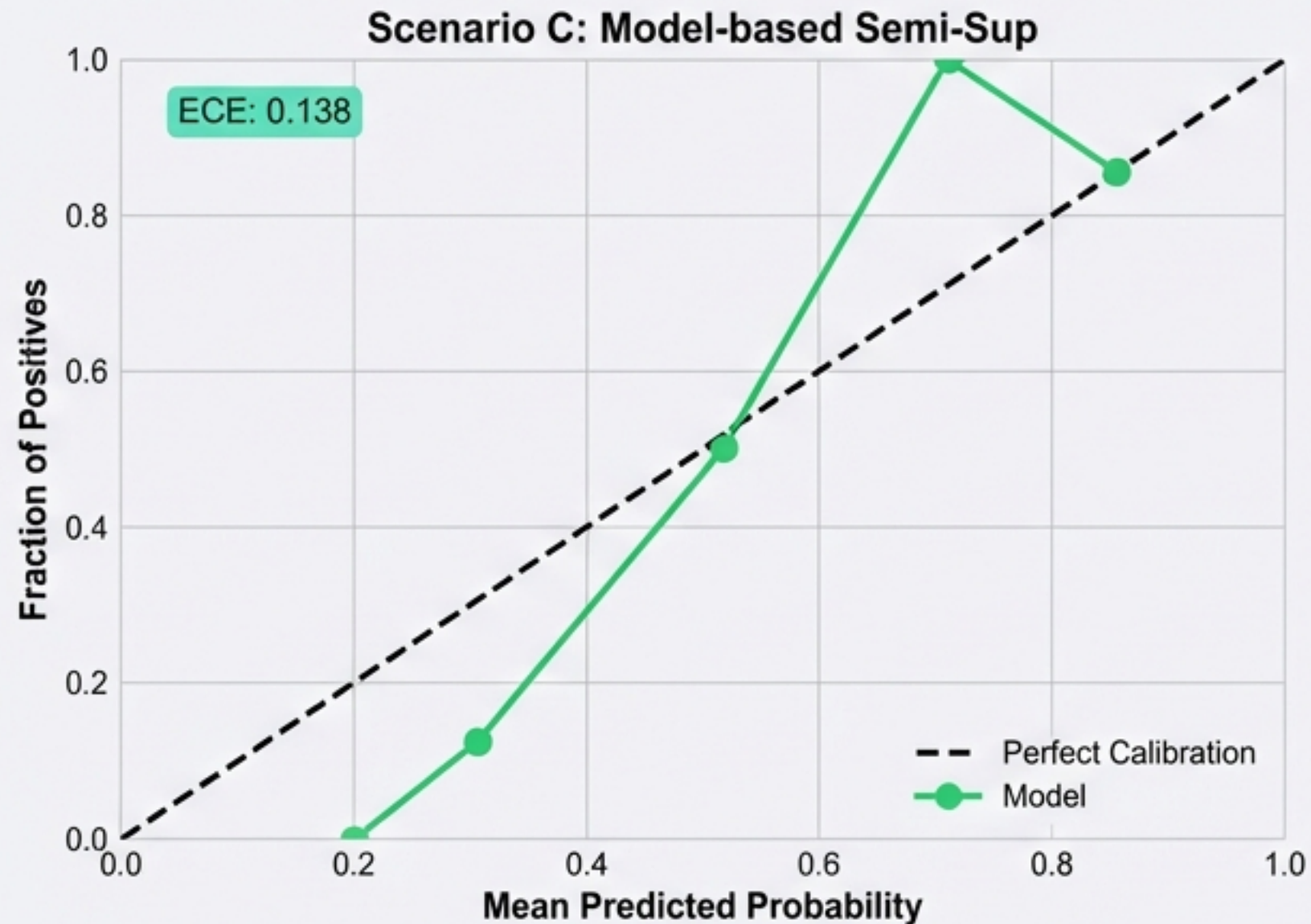
Performance Results: Accuracy vs. Scalability



Scenario A (Supervised)	Scenario B (Clustering)	Scenario C (Self-Training)
F2 Score: 96.43% (± 3.60)	F2 Score: 89.65%	F2 Score: 93.22% (± 4.40)
Recall: 98.00%	Recall: 90.00%	Recall: 94.00%
Result: Best accuracy, but zero scalability.	Result: Performance drop (-7.14%) due to noisy weak labels.	Result: Recovered performance (+3.55% vs B). Difference vs A is not statistically significant (p=0.1237).

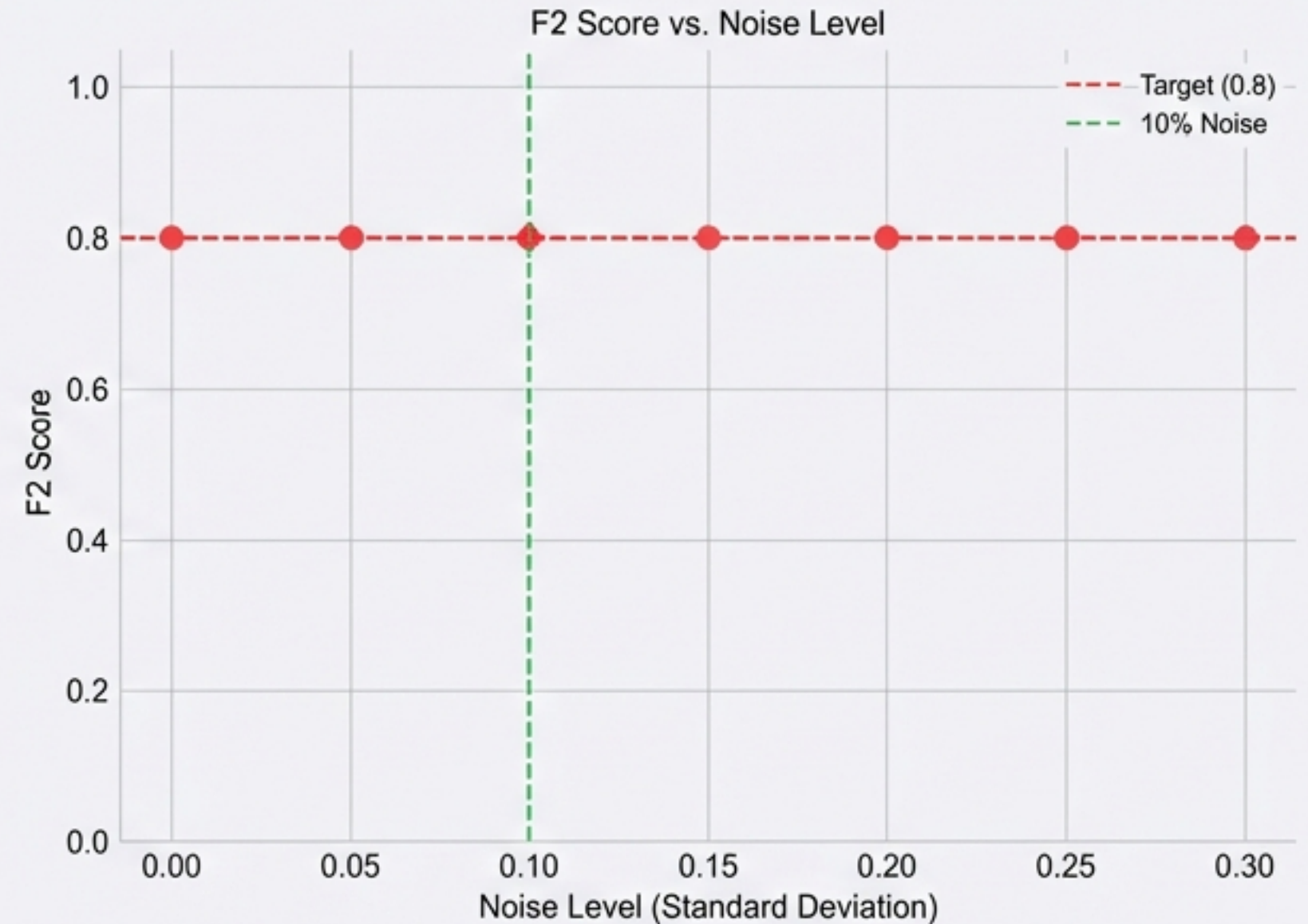
Validation: Robustness & Reliability

Calibration (Trustworthiness)



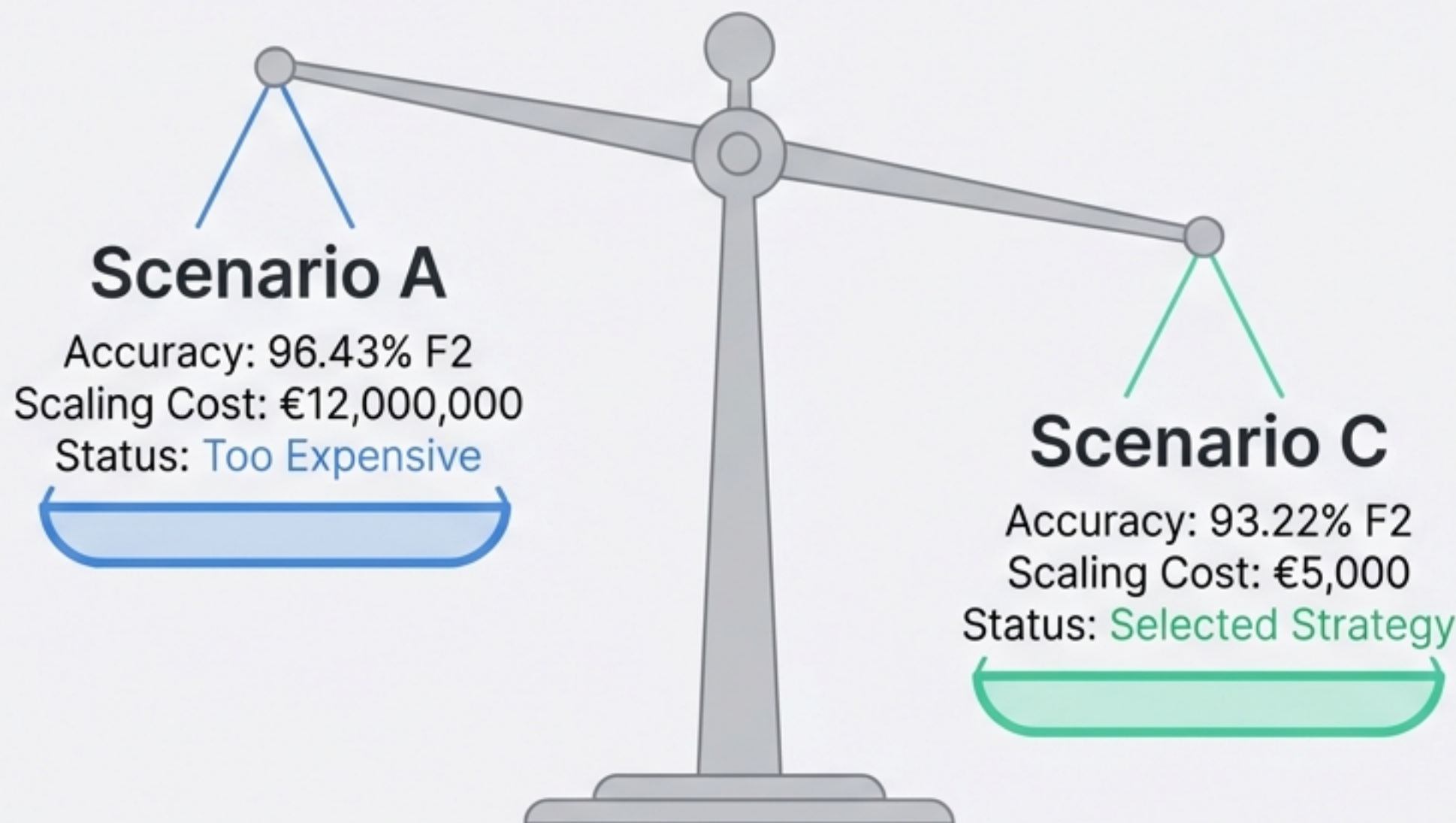
The model's confidence scores align with reality (low Expected Calibration Error), which is critical for the 'Confidence Tiering' strategy in production.

Noise Tolerance (Stability)



When injecting Gaussian noise up to $\pm 10\%$, the F2 Score remained stable at 96.15%. Analysis confirms the model relies on 6 critical PCA components, not random artifacts.

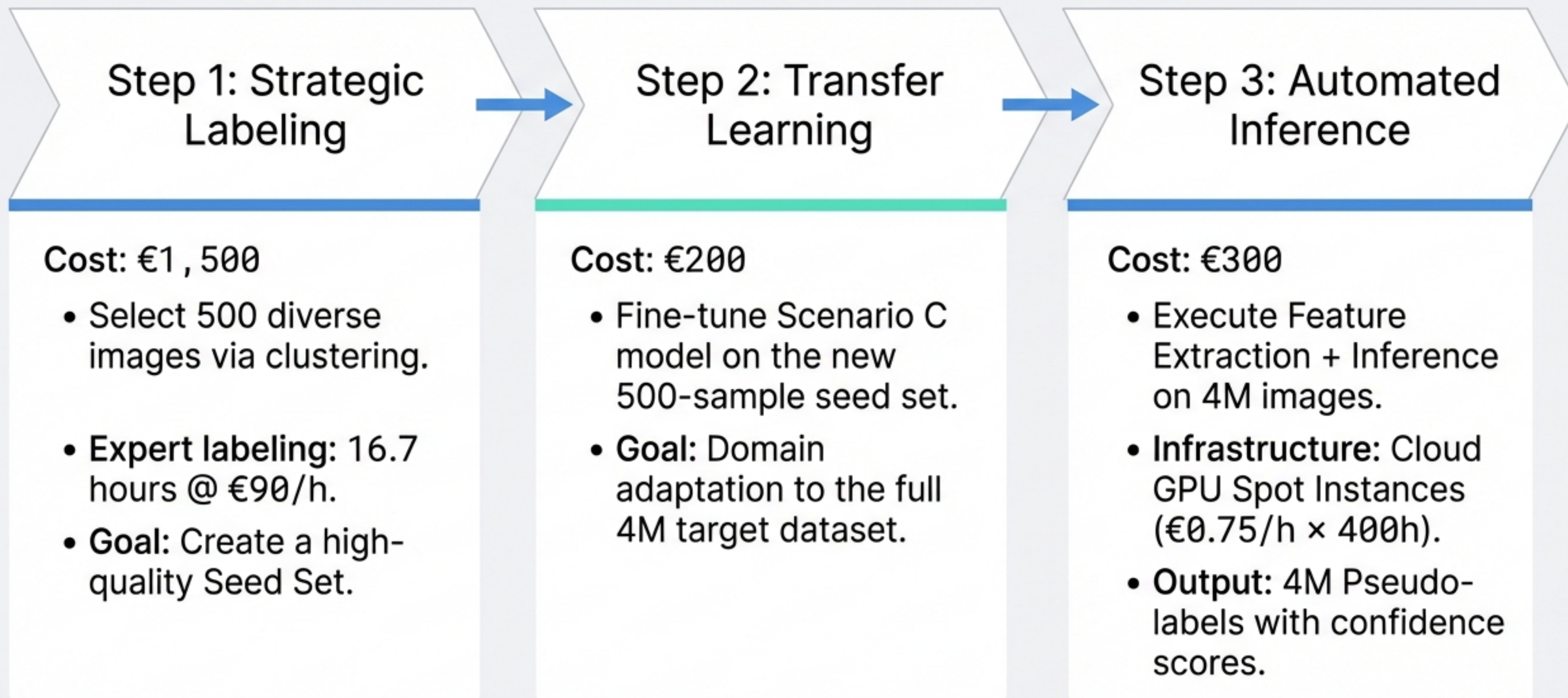
The Strategic Decision: Why Scenario C?



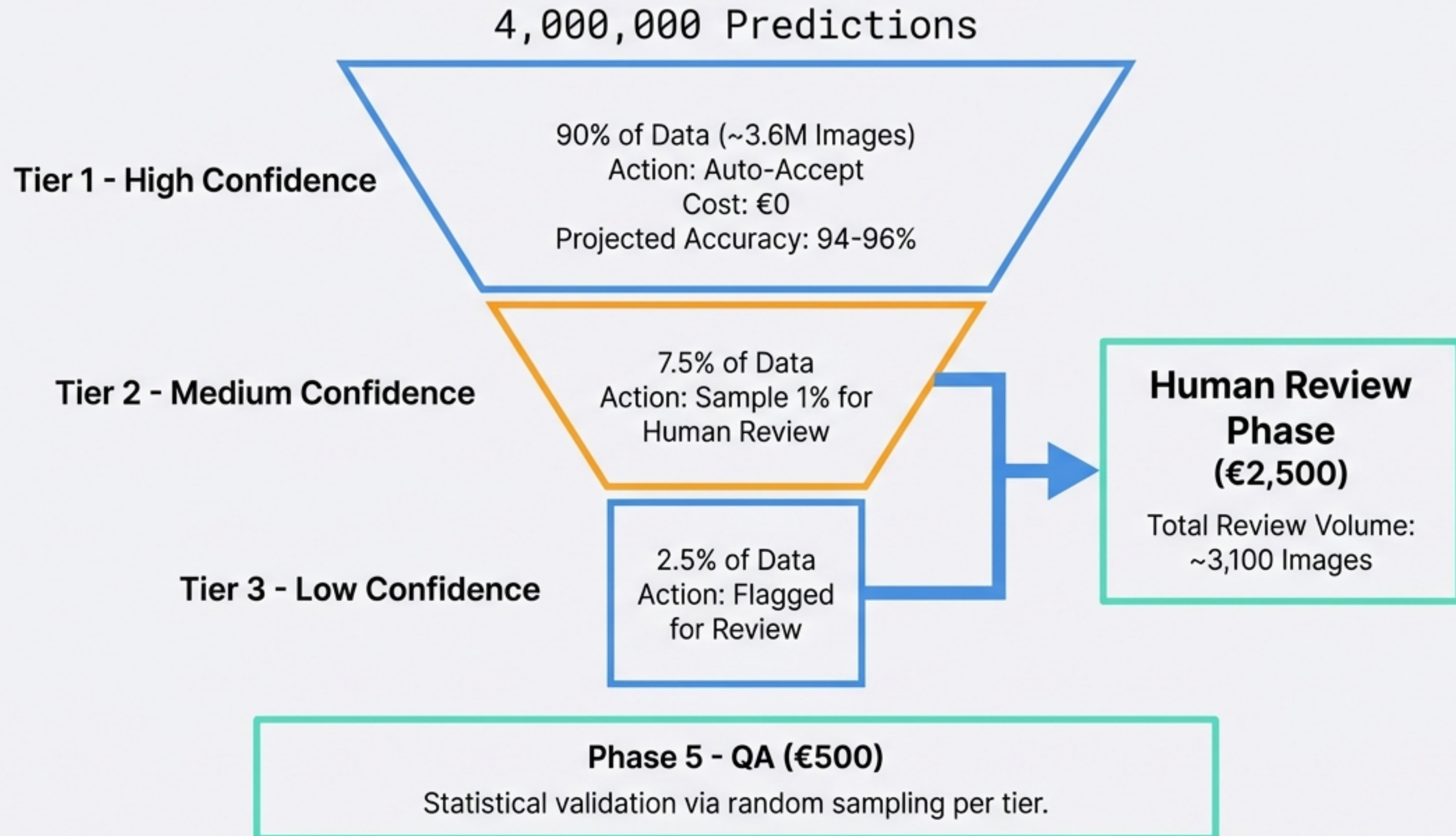
The Trade-off: We accept a -3.21% dip in F2 Score to achieve a 2,400× reduction in cost.

Decision: Deploy Scenario C Methodology. The 93% accuracy is sufficient for pre-screening/research labeling, provided a 'Human-in-the-Loop' mechanism handles low-confidence predictions.

Scaling Strategy: Technical Implementation

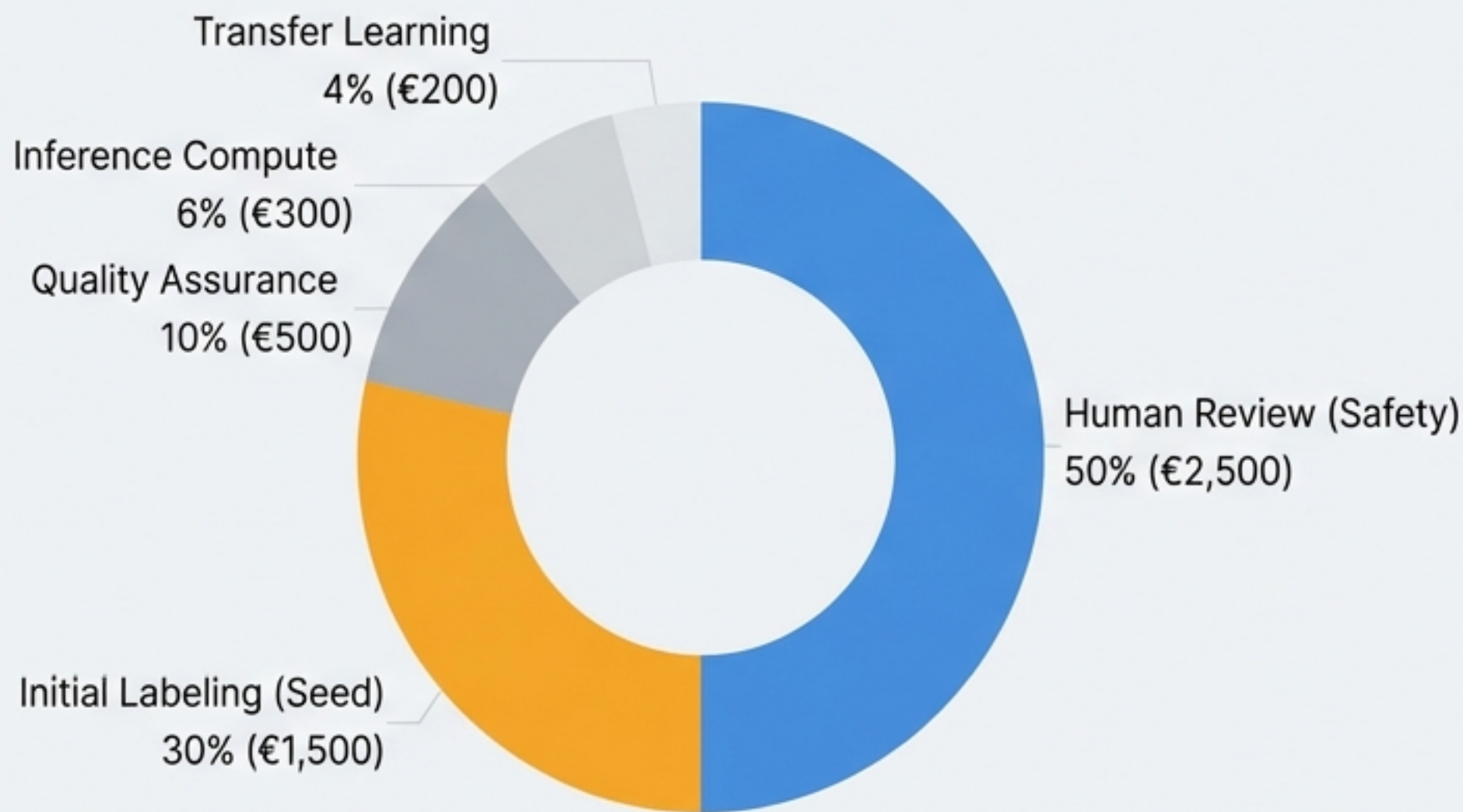


Scaling Strategy: Human-in-the-Loop Quality Control



Budget Allocation & Efficiency

Total Budget: €5,000



Key Statistics

Unit Cost: €0.00125 per image

Total Savings: 99.96% (vs manual baseline)

Strategic Note: The largest spend (50%) is allocated to Safety/Quality Control, not computing power.

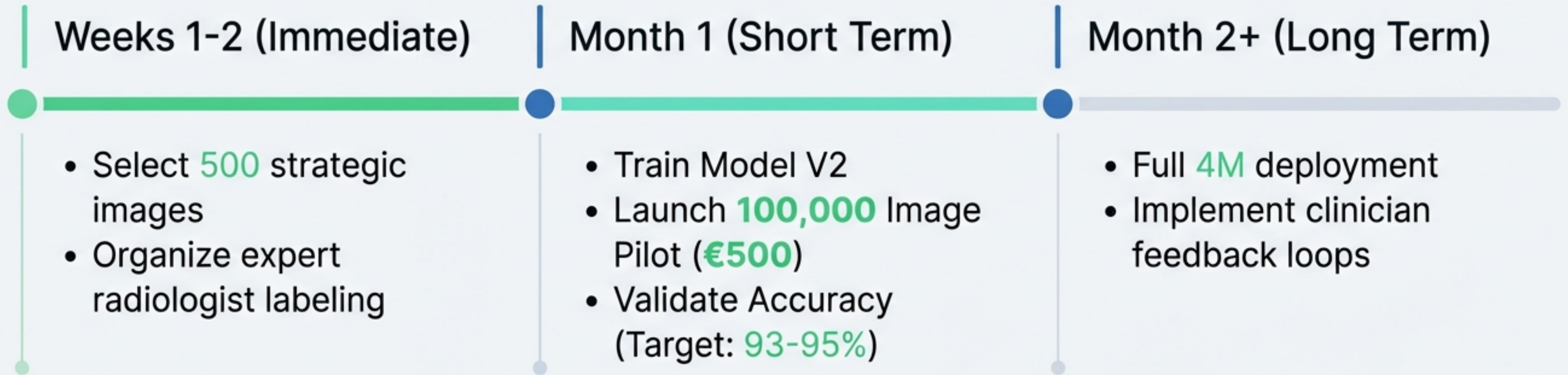
Critical Success Factors & Risk Mitigation

Risk	Impact	Mitigation
Low Pseudo-Label Quality	-7% F2 Score	Aggressive confidence filtering (auto-accept only top tiers).
Distribution Shift	Poor Generalization	External Validation is critical. We will run a Pilot on 100K images first.
GPU Cost Volatility	Budget Overrun	Use Spot Instances (60-70% savings) with strict runtime monitoring.

Prerequisites: Expert validation of initial 500 seed labels and automated pipeline with MLflow tracking.

Conclusion & Roadmap

Feasibility Verdict:  **YES**. Scaling is feasible within budget using Scenario C.



Note: Model is for research screening only, not a certified medical device.